Restaurant Operations & Revenue Analytics

End-to-End SQL + Python + Excel Pipeline

Python | MySQL | Excel

9	7,000+	12	8
Normalized Tables	Transactions Cleaned	SQL Analytical Queries	Excel Visualizations

PROJECT OVERVIEW

Business Problem & Approach

Business Problem

Small independent restaurants manage orders, customers, payments, and deliveries across disconnected systems — making it impossible to reliably analyze revenue trends, peak demand, or operational performance.

Dataset

Synthetic dataset simulating 15 months of restaurant operations (Jan 2025 – Apr 2026). Intentional data quality issues introduced across all 9 tables to simulate real-world messy data.

PIPELINE

01

Database Design

9-table schema
PK / FK constraints

02

Data Generation

7,000+ synthetic orders
Realistic patterns
Intentional messy data

03

Python Cleaning

Deduplicate detection
Type Conversion

04

SQL Analysis

12 analytical queries
JOINs, CASE WHEN
Subqueries

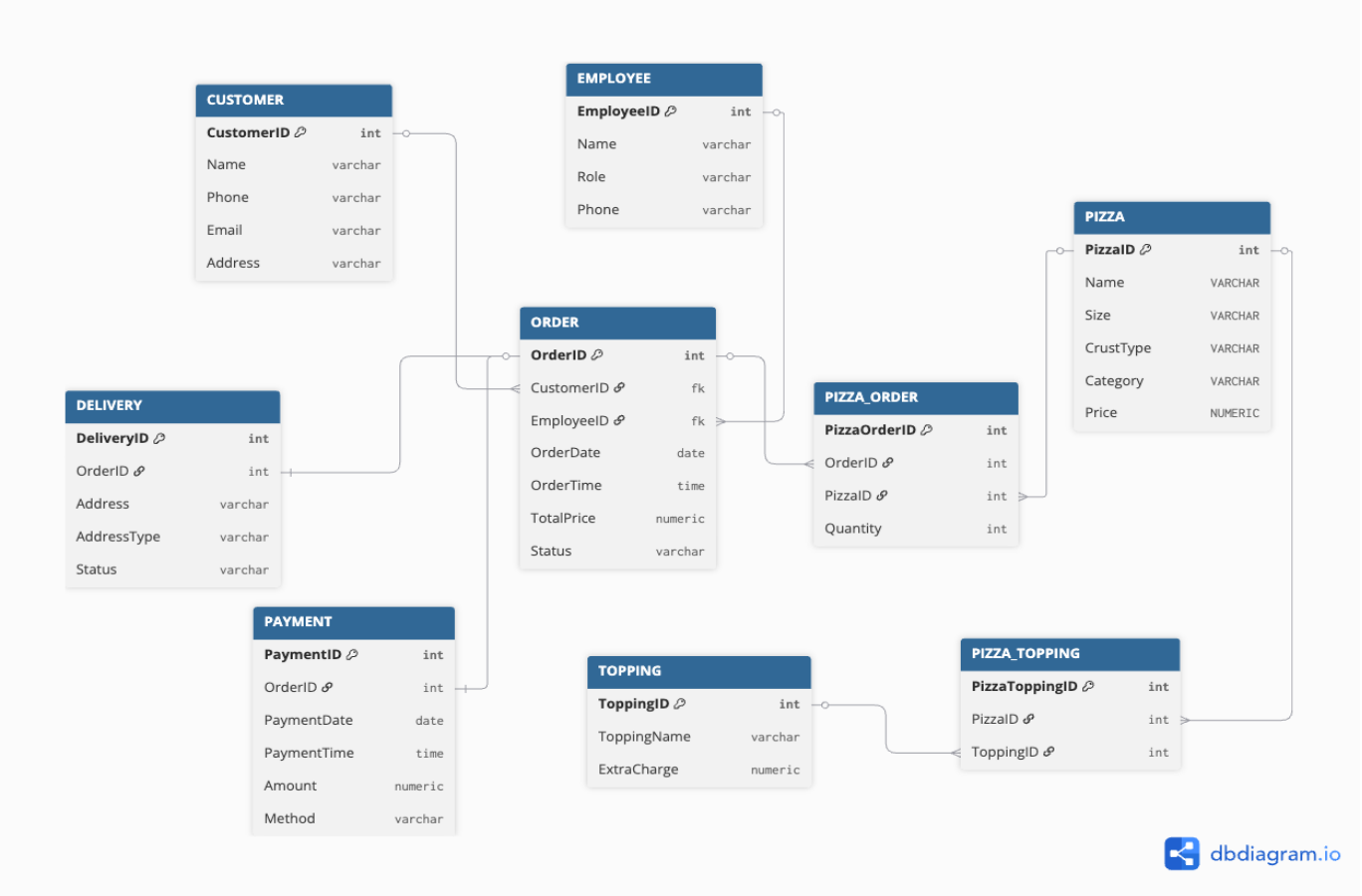
05

Excel Dashboard

8 charts
KPI visualization
Recommendations

DATABASE DESIGN

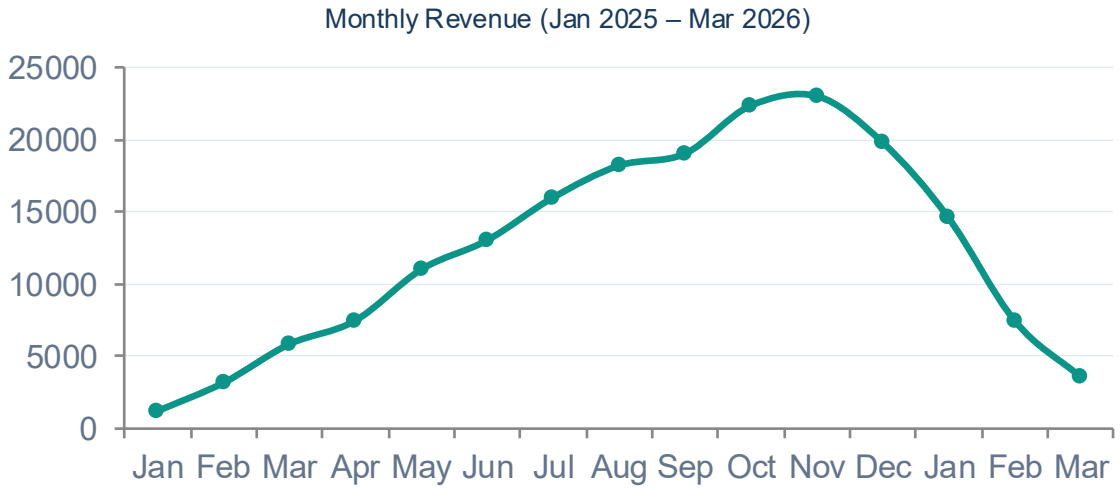
The Schema — What We Built & Why

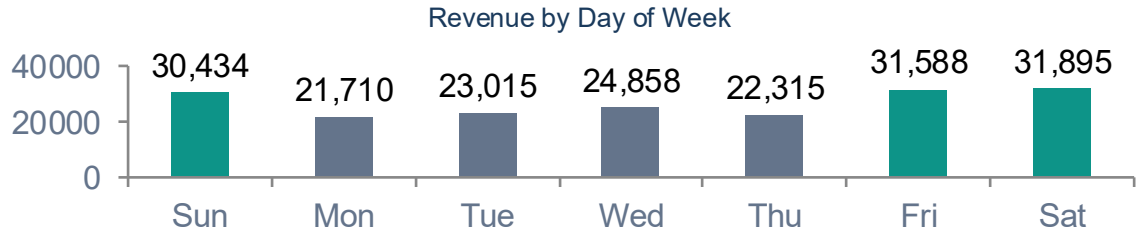


By establishing a connected centralized system for the business, the owner can answer questions that were once unreliable – which pizzas bring in the most revenue, evaluate employee performance, and overall business trends.

REVENUE ANALYSIS

Growth Trend & Demand Patterns





$185,817

Total Revenue
Delivered Orders

20x

Monthly Order Growth
Jan → Nov 2025

$38.58

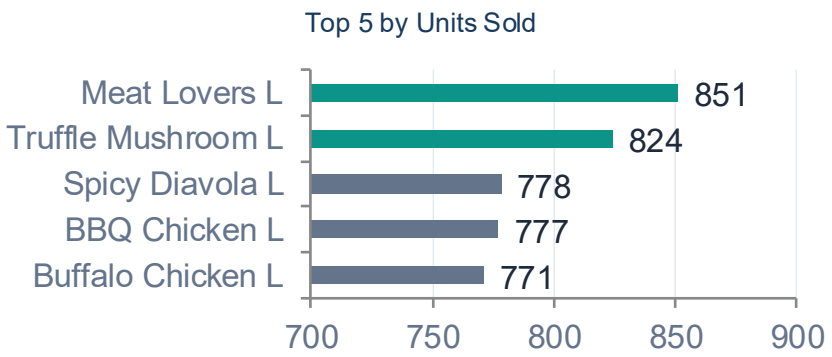
Avg Order Value
Consistent all year

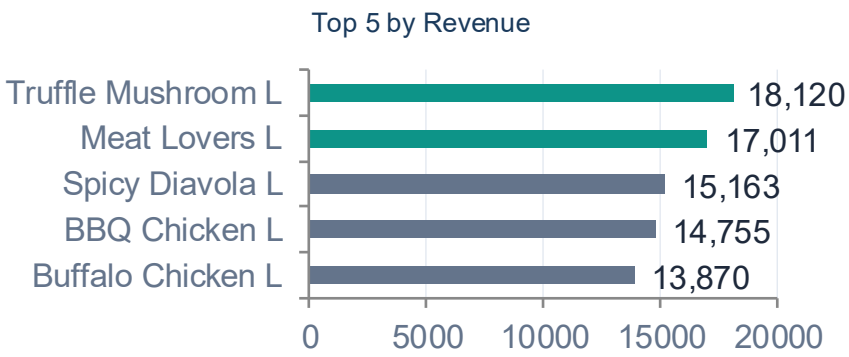
Recommendation

Thursday shows lowest weekday revenue — a targeted promotion could pull forward Friday demand.

MENU PERFORMANCE

Volume vs Revenue — What's Actually Driving the Business





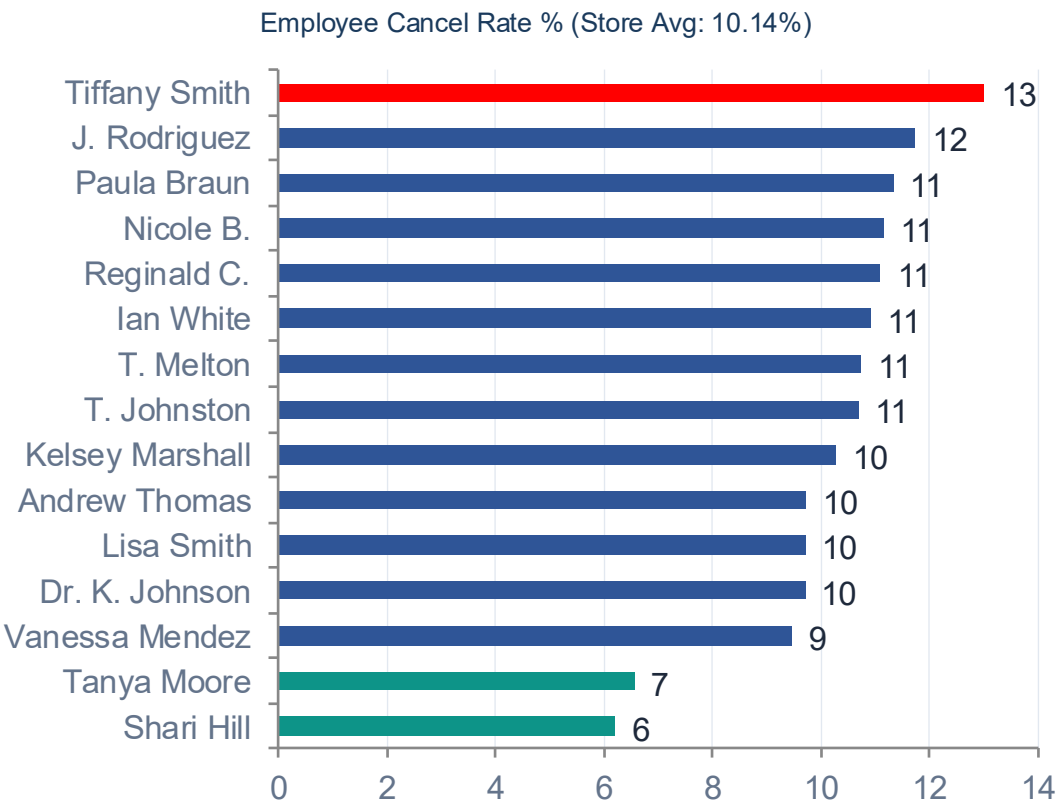
Key Insight — Revenue vs Volume Mismatch

All top 5 pizzas by volume and revenue are Large size – meaning each order uses more ingredients per pizza. Ingredient inventory should account for higher per-pizza consumption, not just order count.

Menu Cut Candidate	Inventory Priority	Pricing Opportunity
Gluten Free Veggie appears in bottom 5 for both volume and revenue across two sizes	All top 5 revenue items are Large size — large size ingredients should always be fully stocked	Modest price increase in Meat Lovers L pizzas could close the gap to Truffle Mushroom L pizzas without sacrificing demand

OPERATIONS & CUSTOMERS

Employee Performance & Customer Value



TOP CUSTOMERS

Jacob Riddle

104 orders · $4,208 spend

Suzanne Soto

98 orders · $3,836 spent

Steven Booth

98 orders · $3,789 spent

Recommendations

- Tiffany Smith cancel rate 28% above store average — management should review performance
- Tanya Moore & Shari Hill at 6.2–6.6% — investigate best practices worth replicating
- 2 of top 10 customers have missing contact info — owner cannot run targeted promotions

WRAP UP

Limitations & Next Steps

LIMITATIONS

Dataset is synthetic — findings are illustrative of methodology, not a real business

No cost data available — profitability and margin analysis not possible

100% repeat customer rate is a data generation artifact — not realistic churn behavior

Shift schedule data absent — employee cancel rate cannot be cross-referenced with staffing

NEXT STEPS

Integrate Cost Data

Enable margin and profitability analysis by menu item

RFM Segmentation

Recency, Frequency, Monetary scoring for customer analytics

Inventory Table

Extend schema to track ingredient usage per order